EE 3072
System Dynamics and Control

Lecture 7
Compensator Design
in Frequency-Domain

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Introduction

Introduction

In frequency-domain the design specifications are usually provided in terms of

- Gain margin, phase margin, and resonant peak magnitude, which are related to the system damping.

- Gain crossover frequency, resonant frequency, and bandwidth, which are related to the speed of the transient response.

- Static error constants, which determine the steady-state accuracy.

Using Bode diagrams it is straightforward to design compensators in order to meet such specifications.

Introduction

In many practical cases compensation is a compromise between steady-state accuracy and relative stability.

Usually it is required that

- The gain in the low frequency region should be sufficiently large.

- Near the gain crossover frequency the slope of the log-magnitude curve should be -20dB/dec.

- This slope should extend on a sufficiently wide frequency range to assure proper phase margin.

- In the high frequency region the gain should be attenuated as fast as possible in order to minimize the effects of noise.

Introduction

Lead compensation improves the transient response, while provides only small change in the steady-state accuracy.

However, it may increase the high-frequency noise effects.

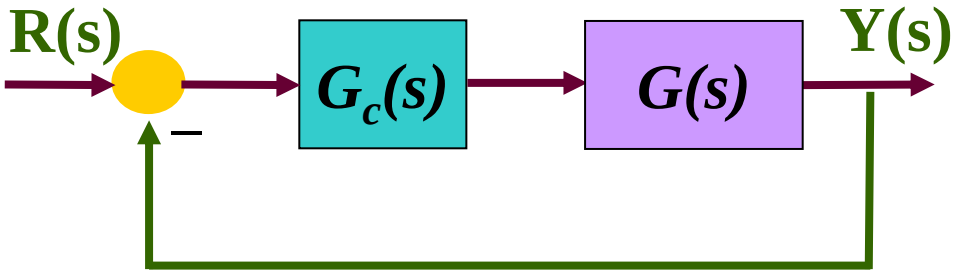
Lag compensation improves the steady-state accuracy at the expense of decreasing the speed of the transient response.

It also attenuates high frequency noise effects.

Lead compensation adds a «derivative» effect, whereas lag compensation adds an «integral» effect to the system.

Phase-Lead Compensator Design

The Phase-Lead Compensator



Recall that the TF of a phase-lead compensator is given by

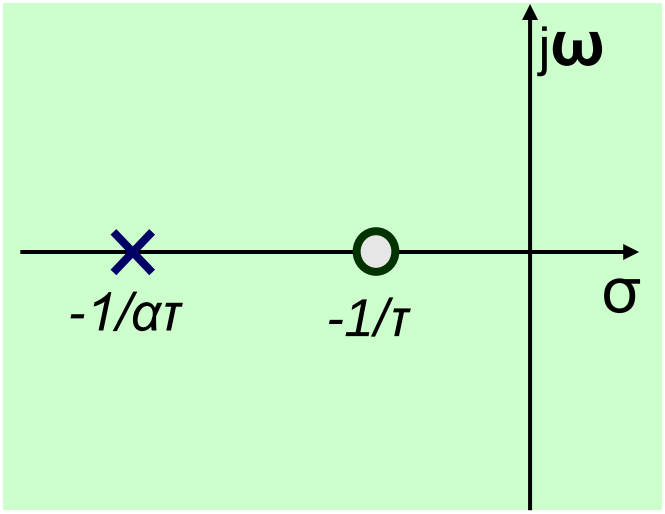
$$G_c(s) = K_c \frac{s + \frac{1}{\tau}}{s + \frac{1}{\alpha\tau}} = K_c\alpha \frac{\tau s + 1}{\alpha\tau s + 1}, 0 < \alpha < 1$$

$$zero: \quad s = -z = -1/\tau$$

$$pole: \quad s = -p = -1/\alpha\tau$$

Its sinusoidal TF is

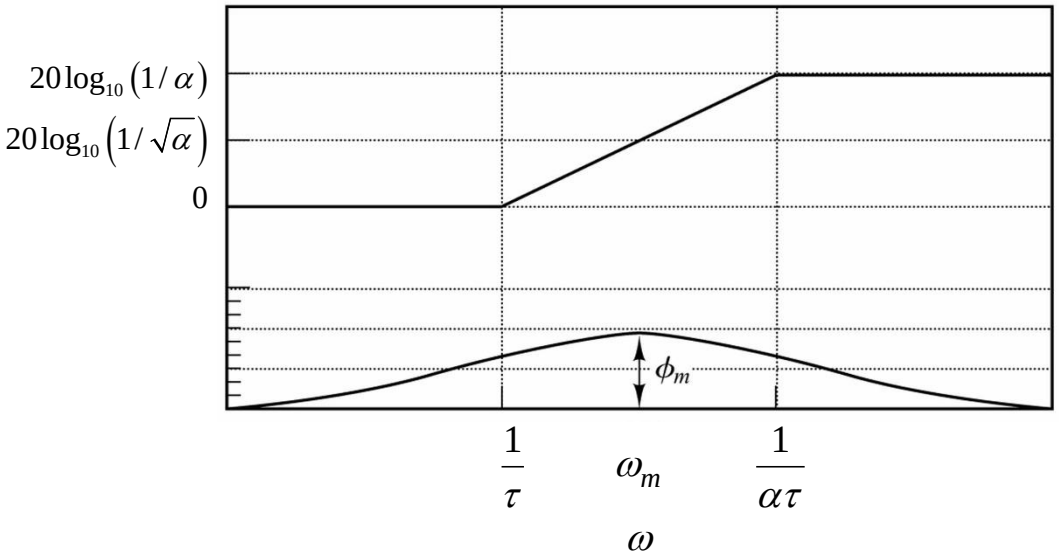
$$G_c(j\omega) = K_c\alpha \frac{1 + j\omega\tau}{1 + j\omega\alpha\tau}, 0 < \alpha < 1$$



The Phase-Lead Compensator

$$G_c(j\omega) = K_c\alpha\frac{1+j\omega\tau}{1+j\omega\alpha\tau}, 0<\alpha<1$$

The Bode diagram of a phase-lead compensator is (assuming $K_c\alpha=1$)



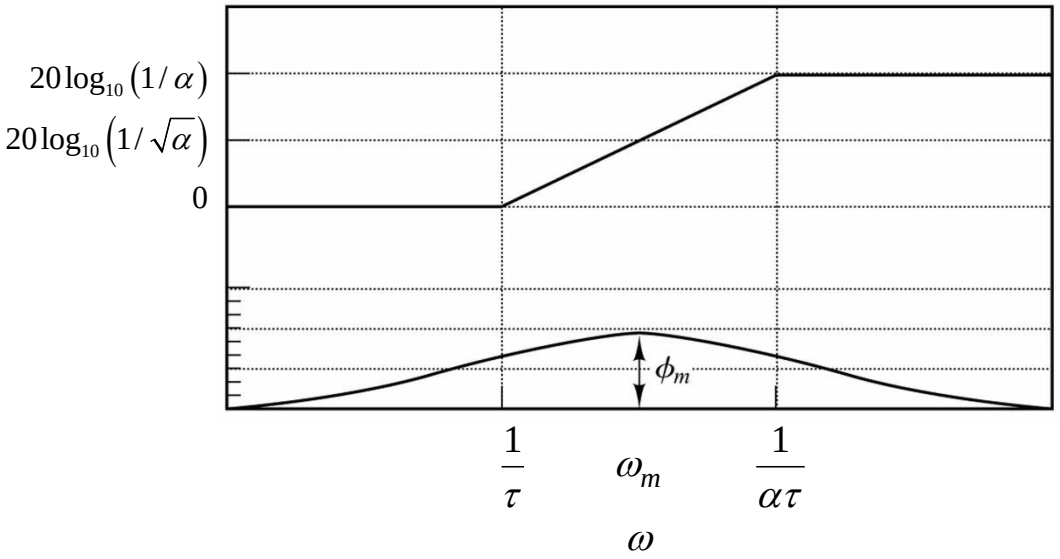
The maximum phase contribution occurs at the frequency ω_m, which is the geometric mean of the corner frequencies.

$$\omega_m = \frac{1}{\tau\sqrt{\alpha}}$$

The maximum phase angle is given by

$$\sin(\phi_m) = \frac{1-\alpha}{1+\alpha}$$

The Phase-Lead Compensator

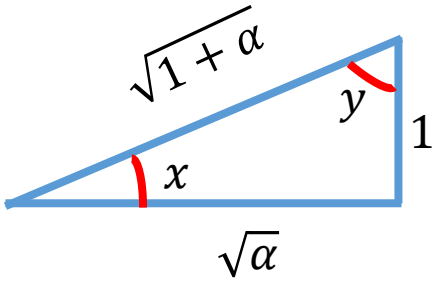


$$G_c(j\omega) = K_c\alpha\frac{1+j\omega\tau}{1+j\omega\alpha\tau},$$

$$0<\alpha<1$$

The maximum phase angle is given by

$$\phi_m = \tan^{-1}\left(\frac{1}{\sqrt{\alpha}}\right) - \tan^{-1}\left(\sqrt{\alpha}\right)$$

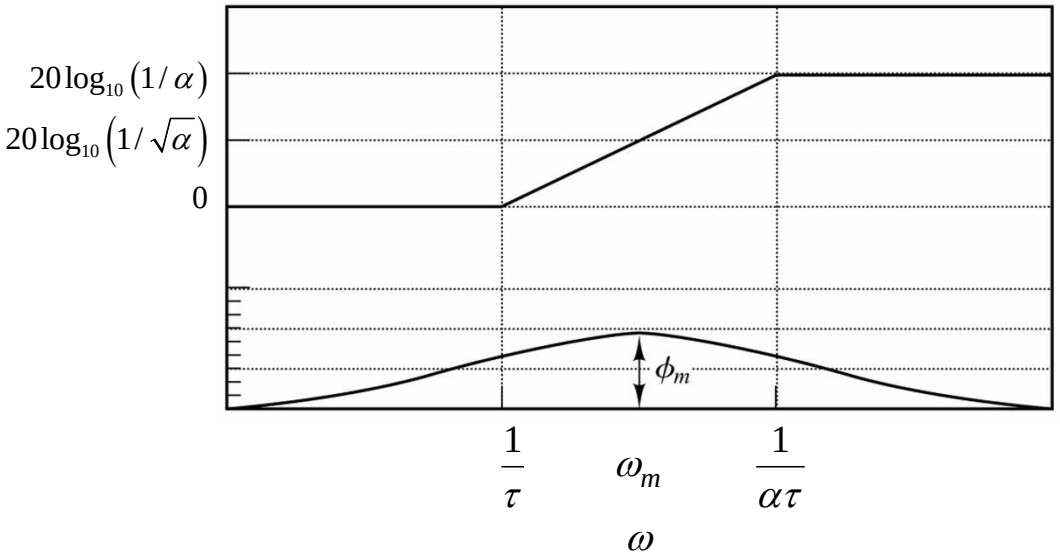


$$\sin\left(\phi_m\right) = \sin\left(x-y\right) = \sin\left(x\right)\cos\left(y\right) - \cos(x)\sin(y)$$

$$\phi_m = x - y$$

$$\sin\left(\phi_m\right) = \left(\frac{1}{\sqrt{1+\alpha}}\right)^2 - \left(\frac{\sqrt{\alpha}}{\sqrt{1+\alpha}}\right)^2 = \frac{1-\alpha}{1+\alpha}$$

The Phase-Lead Compensator



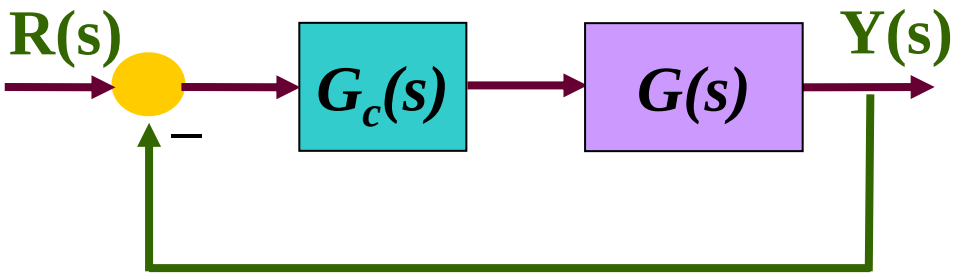
$$G_c(j\omega) = K_c\alpha\frac{1+j\omega\tau}{1+j\omega\alpha\tau},$$

$$0 < \alpha < 1$$

A phase-lead compensator adds a positive phase to (i.e., phase lead) to the system – hence the name.

It is a high-pass filter (attenuating low frequencies and allowing high frequencies to pass).

Phase-Lead Compensator Design Procedure



1. Assume that the compensator TF has the from

$$G_c(s) = K_c \frac{s+1/\tau}{s+1/\alpha\tau} = \underbrace{K_c\alpha}_{K} \frac{\tau s+1}{\alpha\tau s+1} = K\frac{\tau s+1}{\alpha\tau s+1}, 0<\alpha<1$$

Let $K_c\alpha=K$ and determine the value of K to satisfy the steady-state error requirements.

2. Using the determined value of K sketch the Bode diagram of the (gain adjusted) uncompensated system

$$G_1(s) = KG(s)$$

and calculate its phase margin.

Phase-Lead Compensator Design Procedure

3. From the desired and actual phase margins determine the necessary phase-lead allowing for small amount of safety

$$\phi_m = \underset{\text{desired}}{\phi_d} - \underset{\text{actual}}{\phi_a} + \underset{\text{safety}}{\varepsilon}$$

ϕ_d is the desired phase margin for the compensated system.

ϕ_a is the actual phase margin for the uncompensated system.

ε is small angle added for safety (to compensate for the ignored nonidealities)

ϕ_m must be provided by the compensator.

Phase-Lead Compensator Design Procedure

4. From

$$\sin\left(\phi_m\right)=\frac{1-\alpha}{1+\alpha}$$

determine the value of α.

Also, from the magnitude curve of $G_1(j\omega)$ determine the value of the frequency at which the magnitude is equal to

$$-20\log_{10}\left(1/\sqrt{\alpha}\right)$$

Select this frequency as the new gain crossover frequency. It corresponds to

$$\omega_m=\frac{1}{\tau\sqrt{\alpha}}$$

at which the maximum phase shift ϕ_m occurs. Determine the value of τ from ω_m.

Phase-Lead Compensator Design Procedure

5. Calculate the corner frequencies

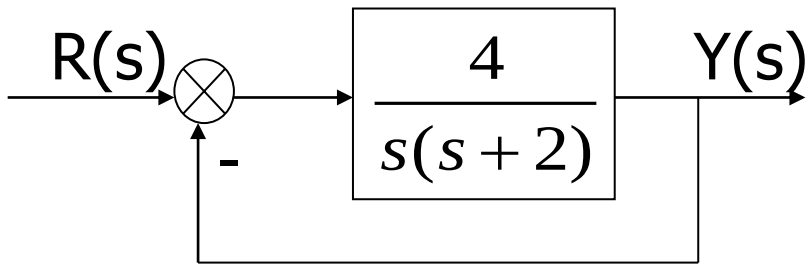
$$z = 1/\tau \quad \text{(the zero)} \quad \text{and} \quad p = 1/\alpha\tau \quad \text{(the pole)}$$

6. Determine the value of K_c from $K_c = K/\alpha$.

7. Check the phase margin of the compensated system to make sure that it is satisfactory. If not, repeat the procedure by modifying the pole-zero locations.

Phase-Lead Compensator Design

Example: Consider the system



Open-loop TF: $G(s) = \dfrac{4}{s(s+2)}$

Closed-loop TF: $T(s) = \dfrac{4}{s^2+2s+4}$

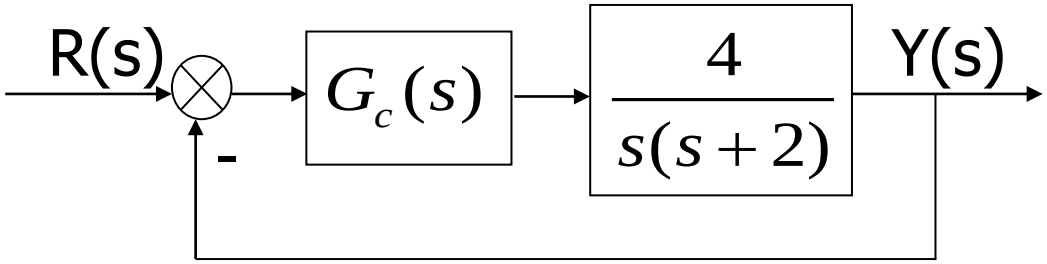
Design specifications:

It is desired to design a compensator so that the steady-state error to a ramp input is less than 5%, and the phase margin of the CL system is at least 50°.

Solution. Choose a phase-lead compensator.

$$G_c(s) = K_c \frac{s+1/\tau}{s+1/\alpha\tau} = \underbrace{K_c\alpha}_{K} \frac{\tau s+1}{\alpha\tau s+1} = K\frac{\tau s+1}{\alpha\tau s+1}, 0<\alpha<1$$

Phase-Lead Compensator Design



$$G_c(s) = K\frac{\tau s+1}{\alpha\tau s+1},$$

$$0<\alpha<1, \quad K = K_c\alpha$$

The steady-state error for a ramp input is

$$e_{ss} = \frac{1}{K_v} \le 0.05 \quad \Rightarrow \quad K_v \ge 20 \quad \Rightarrow \quad \text{Let} \quad K_v = 20$$

The velocity error constant for the CL system is

$$K_v = \lim_{s\to 0} sG_c(s)G(s) = \lim_{s\to 0} sK\frac{(\tau s+1)}{(\alpha\tau s+1)}\frac{4}{s(s+2)} = 2K = 20 \Rightarrow K = 10$$

Draw the Bode diagram of

$$G_1(s) = KG(s) = \frac{40}{s(s+2)} \Rightarrow$$

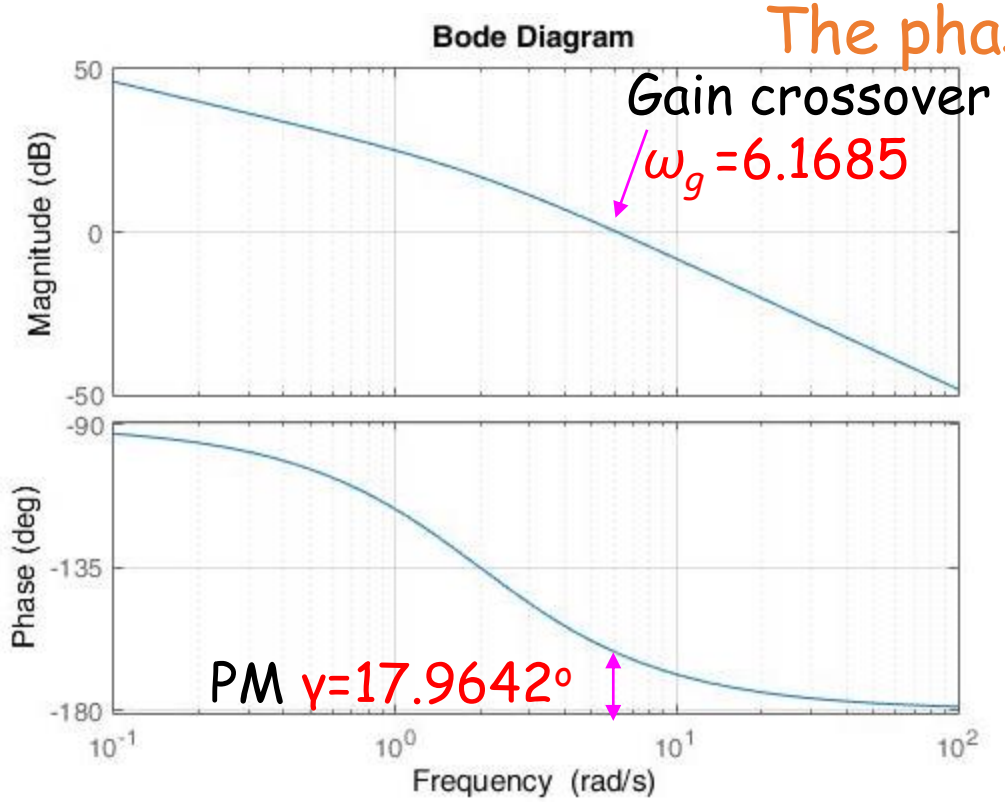
$$G_1(j\omega) = \frac{40}{(j\omega)(j\omega+2)}$$

$$= \frac{20}{(j\omega)(1+j\omega/2)}$$

Phase-Lead Compensator Design

Draw the Bode diagram of

$$G_1(j\omega) = \frac{20}{(j\omega)(1+j\omega/2)}$$



The gain crossover frequency is

$$\omega\sqrt{1+\frac{\omega^2}{4}} = 20 \Rightarrow \omega_g = 6.1685$$

The phase angle of $G_1(j\omega)$ at $\omega=\omega_g$ is

$$\phi = -90^o - \tan^{-1}\left(\frac{6.1685}{2}\right)$$

$$= -162.0358^o$$

The phase margin is

$$\gamma = 180^o - 162.0358^o = 17.9642^o$$

The phase angle to be provided by the compensator is

$$\phi_m = \underset{\text{desired}}{50^o} - \underbrace{17.9642^o}_{\text{actual}} + \underset{\text{safety}}{5^o} = 37.0358^o$$

Phase-Lead Compensator Design

Determine α

$$\sin\left(37.0358^{o}\right)=\frac{1-\alpha}{1+\alpha}=0.6023$$

$$\alpha=0.2482$$

Determine ω_m

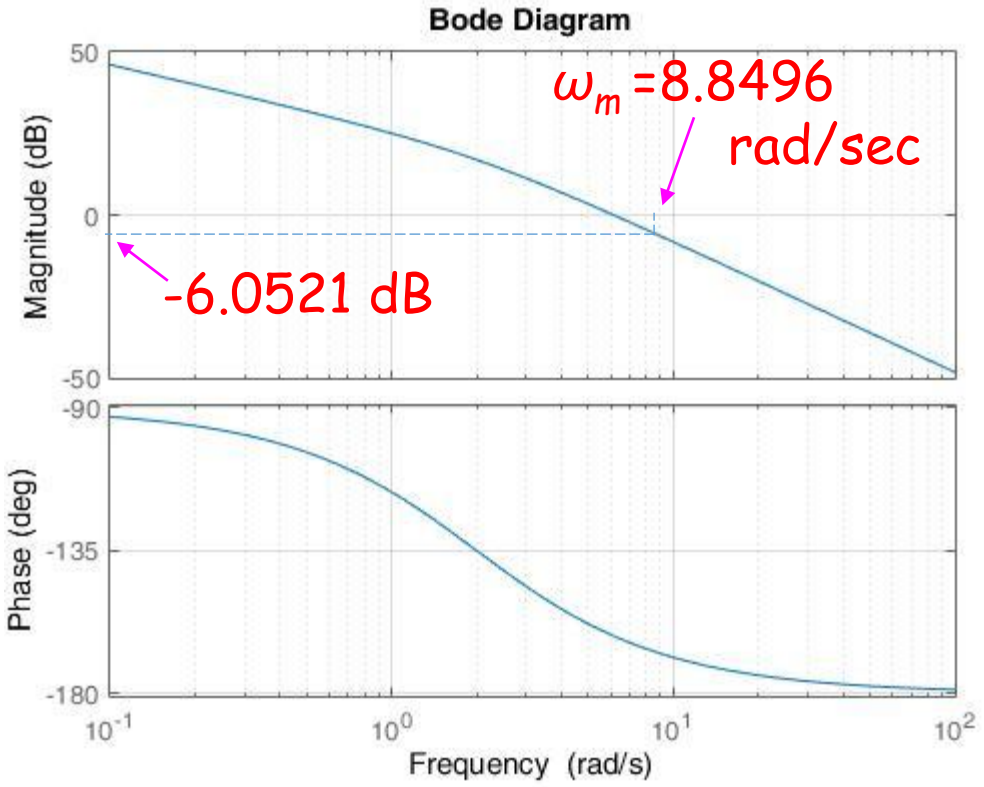
$$-20\log_{10}\left(1/\sqrt{0.2482}\right)=-6.0521$$

$$20\log_{10}\left(\left|G_1(j\omega_m)\right|\right)=-6.0521$$

$$20\log_{10}\left(\frac{20}{\omega_m\sqrt{1+\omega_m^2/4}}\right)=-6.0521$$

$$\omega_m=8.8496\ \text{rad/sec}$$

ω_m will be the new corner frequency of the compensated system



Phase-Lead Compensator Design

Determine the pole and the zero

$$\frac{1}{\tau} = \omega_m \sqrt{\alpha} = 8.8496\sqrt{0.2482} = 4.4088 = z$$

$$\frac{1}{\alpha\tau} = \frac{4.4088}{0.2482} = 17.7633 = p$$

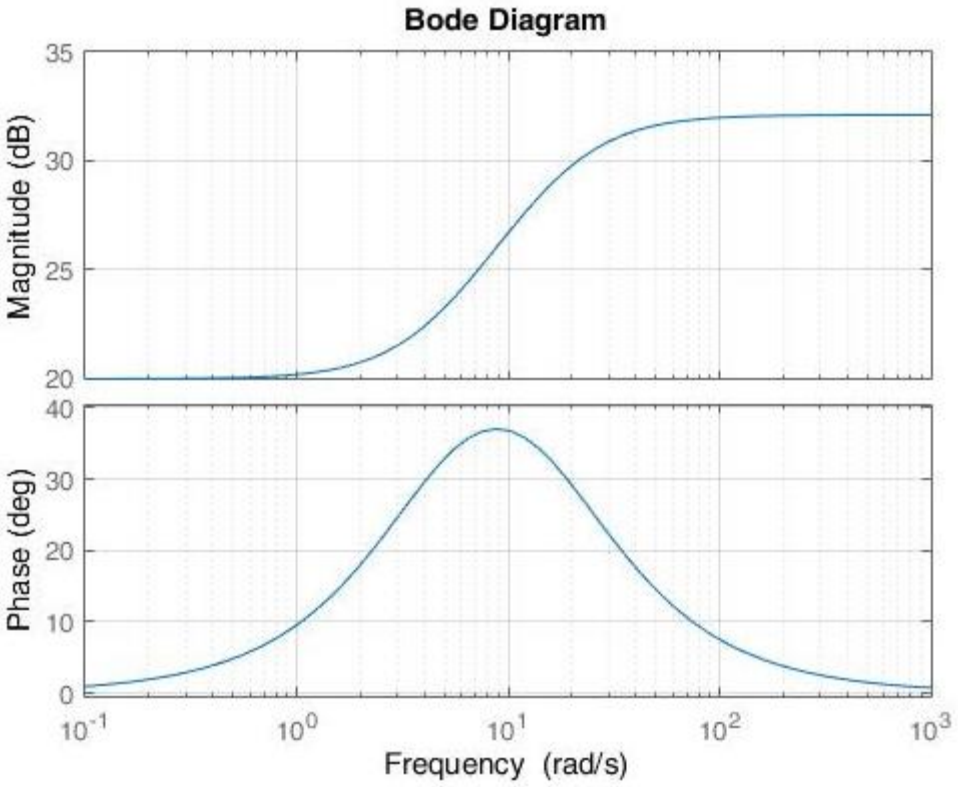
Determine the gain K_c

$$K_c = \frac{K}{\alpha} = \frac{10}{0.2482} = 40.2901$$

The compensator TF is

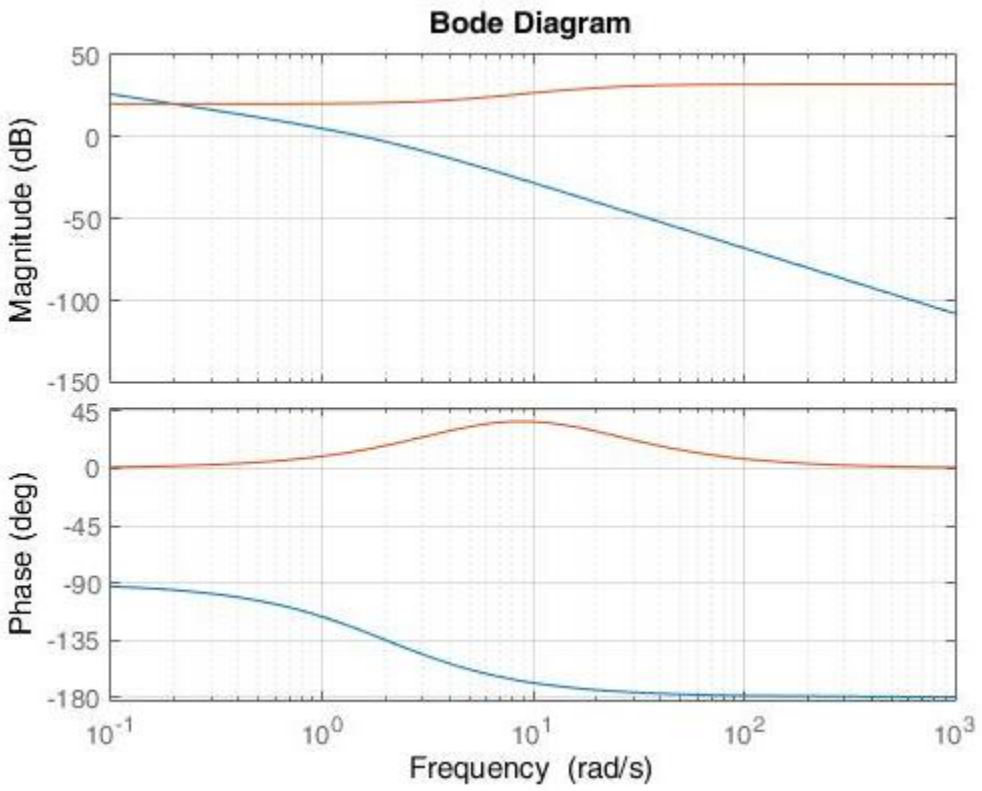
$$G_c(s) = 40.2901 \frac{s + 4.4088}{s + 17.7633}$$

The Bode diagram of the compensator is



Phase-Lead Compensator Design

The Bode diagrams of the plant and the compensator are



The compensator shifts up the magnitude curve.

It moves the gain crossover frequency to the right improving the bandwidth of the system.

It adds positive phase to the system.

The maximum phase is added around the new crossover frequency improving the phase margin.

Phase-Lead Compensator Design

The OL TF of the compensated system is

$$G_c(s)G(s) = 40.2901\frac{(s+4.4088)}{(s+17.7633)}\frac{4}{s(s+2)}$$

It has the velocity error constant

$$K_v = 19.9998 \cong 20$$

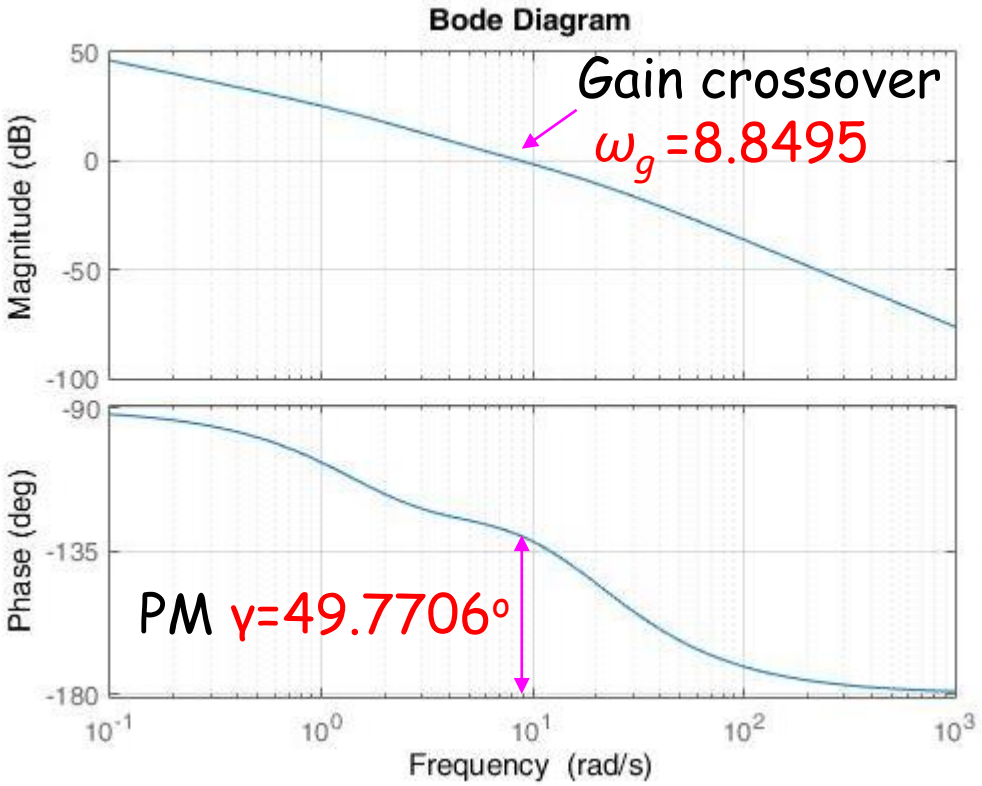
The new gain crossover frequency is ω_g=8.8495 rad/sec.

The phase at $\omega=\omega_g$ is

$$\measuredangle G_c(j\omega_g)G(j\omega_g) = -90^o - \tan^{-1}\left(\frac{8.8495}{2}\right)$$

$$+\tan^{-1}\left(\frac{8.8495}{4.4088}\right) - \tan^{-1}\left(\frac{8.8495}{17.7633}\right) = -130.2294^o$$

The Bode diagram of the compensated system is



The new phase margin is

$$\gamma = 180^o - 130.2294^o = 49.7706^o$$

Phase-Lead Compensator Design

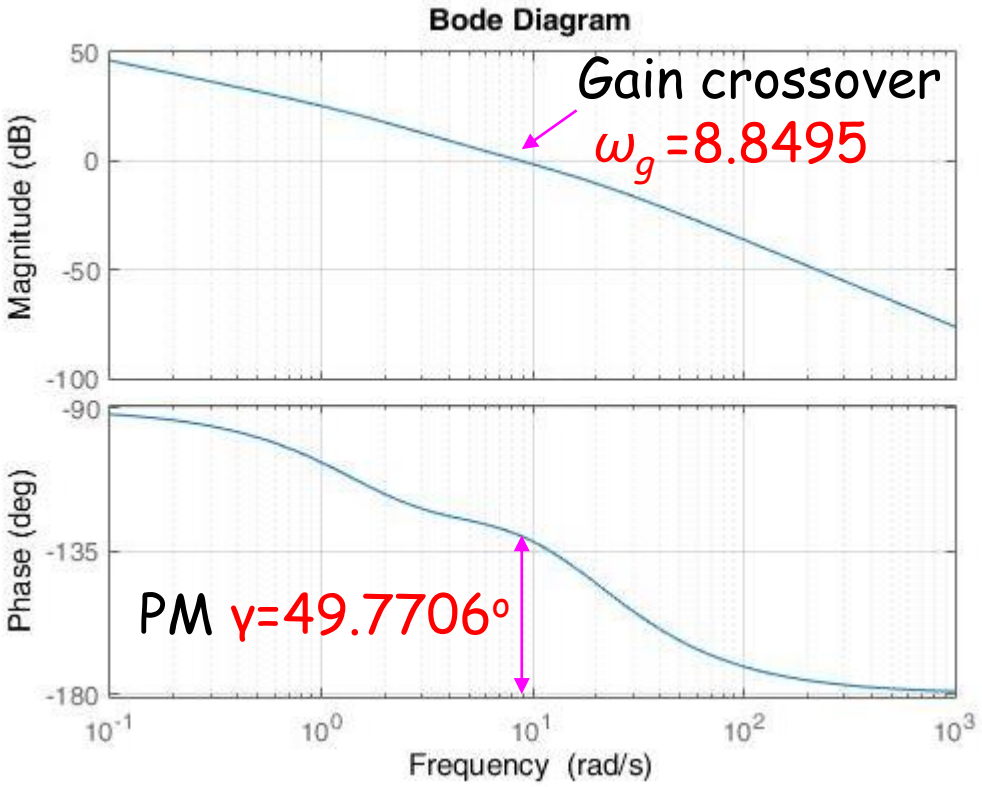
The OL TF of the compensated system is

$$G_c(s) = 40.2901 \frac{(s+4.4088)}{(s+17.7633)} \frac{4}{s(s+2)}$$

Note that for this system the gain margin is ∞.

Also, with compensation the bandwidth of the system has been increased (the gain crossover frequency increased from 6.1685 to 8.8495 and similarly the bandwidth).

The Bode diagram of the compensated system is

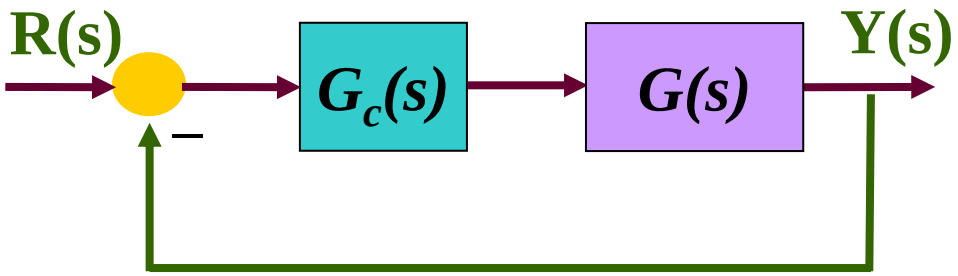


=> increase in the speed of the response is achieved.

Since the specifications are satisfied, the design is completed.

Phase-Lag Compensator Design

The Phase-Lag Compensator

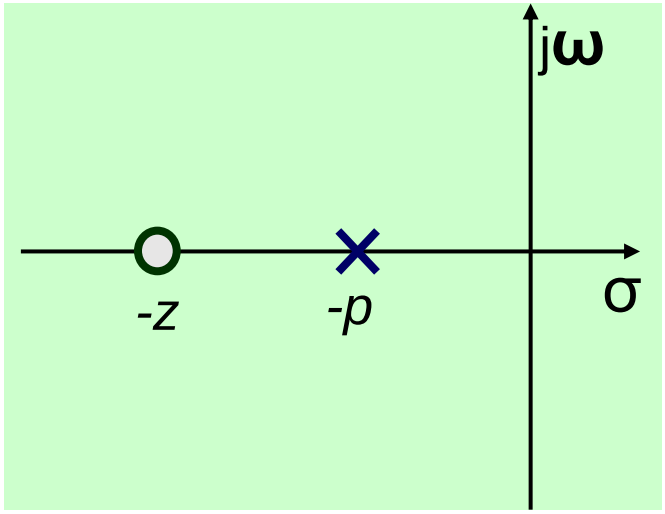


Recall that the TF of a phase lag compensator is given by

$$G_c(s) = K_c \frac{s + \frac{1}{\tau}}{s + \frac{1}{\beta\tau}} = K_c\beta \frac{\tau s + 1}{\beta\tau s + 1}, \beta > 1$$

$$zero: \quad s = -z = -1/\tau$$

$$pole: \quad s = -p = -1/\beta\tau$$

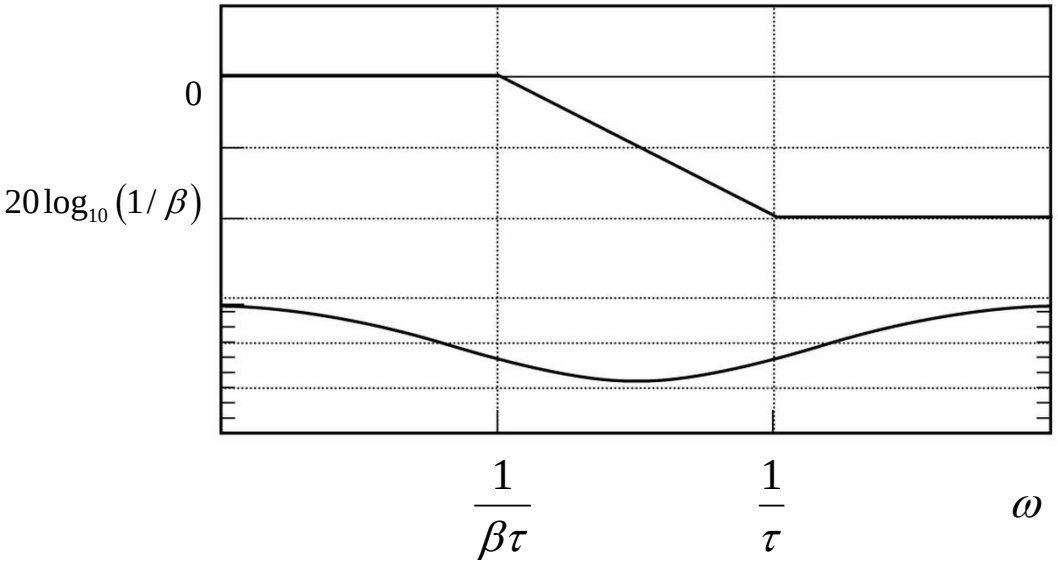


Its sinusoidal TF is

$$G_c(j\omega) = K_c\beta \frac{1 + j\omega\tau}{1 + j\omega\beta\tau}, \beta > 1$$

The Phase-Lag Compensator

The Bode diagram of a phase-lag compensator is (assuming $K_c\beta=1$)



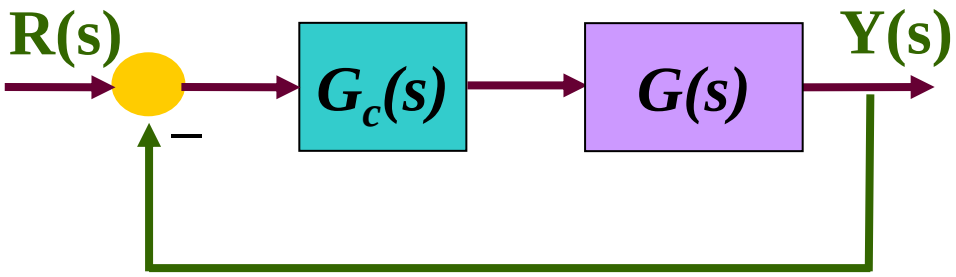
$$G_c(j\omega) = K_c\beta\frac{1+j\omega\tau}{1+j\omega\beta\tau}, \beta > 1$$

Note that the phase-lag compensator is a low-pass filter.

The phase-lag compensator adds a negative phase (phase lag) to the system – hence the name phase lag.

It provides attenuation in the high frequency range. Therefore, it lowers the 0-dB crossover frequency of the system. This increases the phase margin of the system since at lower frequencies we usually have larger phase margin.

Phase-Lag Compensator Design Procedure



1. Assume that the compensator TF has the from

$$G_c(s) = K_c \frac{s+1/\tau}{s+1/\beta\tau} = \underbrace{K_c\beta}_{K} \frac{\tau s+1}{\beta\tau s+1} = K\frac{\tau s+1}{\beta\tau s+1}, \beta > 1$$

Let $K_c\beta=K$ and determine the value of K to satisfy the steady-state error requirements.

Phase-Lag Compensator Design Procedure

2. Using the determined value of K sketch the Bode diagram of the (gain adjusted) uncompensated system

$$G_1(s) = KG(s)$$

and calculate its phase margin.

If it is insufficient, then determine the frequency at which the phase margin is satisfied, i.e., the frequency at which the phase angle is equal to

$$\phi = \measuredangle G_1(j\omega) = -180^o + \phi_d + \varepsilon$$

Choose this frequency as the new gain crossover frequency.

Phase-Lag Compensator Design Procedure

3. Place the zero (choose the corner frequency)

$z = 1/\tau$ (the zero) $\omega = 1/\tau$ (the corner frequency)

one decade below the new crossover frequency.

4. Determine the necessary attenuation to bring the curve to 0-dB at the new gain crossover frequency. It must be equal to

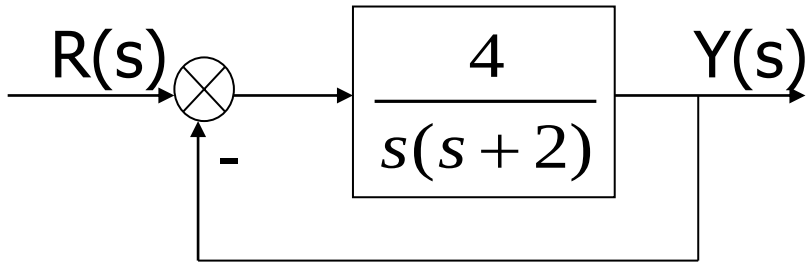
$$-20\log_{10}(1/\beta)$$

Then, determine the value of β and the corner frequency of the pole $p=1/\beta\tau$.

5. Determine the value of K_c from $K_c=K/\beta$.

Phase-Lag Compensator Design

Example: Consider again the system



Open-loop TF: $G(s) = \dfrac{4}{s(s+2)}$

Closed-loop TF: $T(s) = \dfrac{4}{s^2+2s+4}$

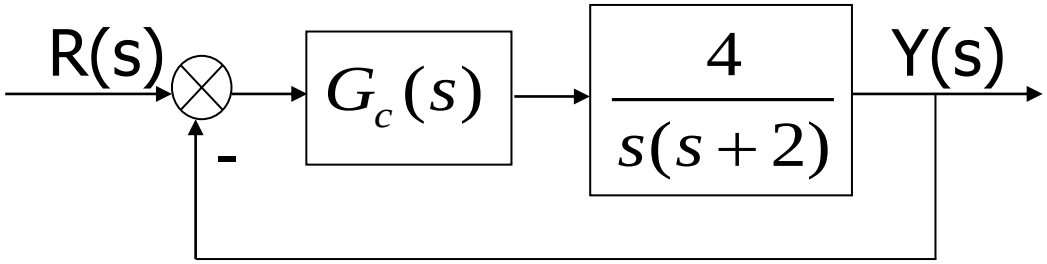
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It is desired to design a compensator so that the steady-state error to a ramp input is less than 5%, and the phase margin of the CL system is at least 50°.

Solution. Choose a phase-lag compensator.

$$G_c(s) = K_c \frac{s+1/\tau}{s+1/\beta\tau} = \underset{K}{K_c\beta} \frac{\tau s+1}{\beta\tau s+1} = K\frac{\tau s+1}{\beta\tau s+1}, \beta > 1$$

Phase-Lag Compensator Design



$$G_c(s) = K\frac{\tau s+1}{\beta\tau s+1},$$

$$\beta > 1, \quad K = K_c\beta$$

The steady-state error for a ramp input is

$$e_{ss} = \frac{1}{K_v} \le 0.05 \quad \Rightarrow \quad K_v \ge 20 \quad \Rightarrow \quad \text{Let} \quad K_v = 20$$

The velocity error constant for the CL system is

$$K_v = \lim_{s\to 0} sG_c(s)G(s) = \lim_{s\to 0} sK\frac{(\tau s+1)}{(\beta\tau s+1)}\frac{4}{s(s+2)} = 2K = 20 \Rightarrow K = 10$$

Draw the Bode diagram of

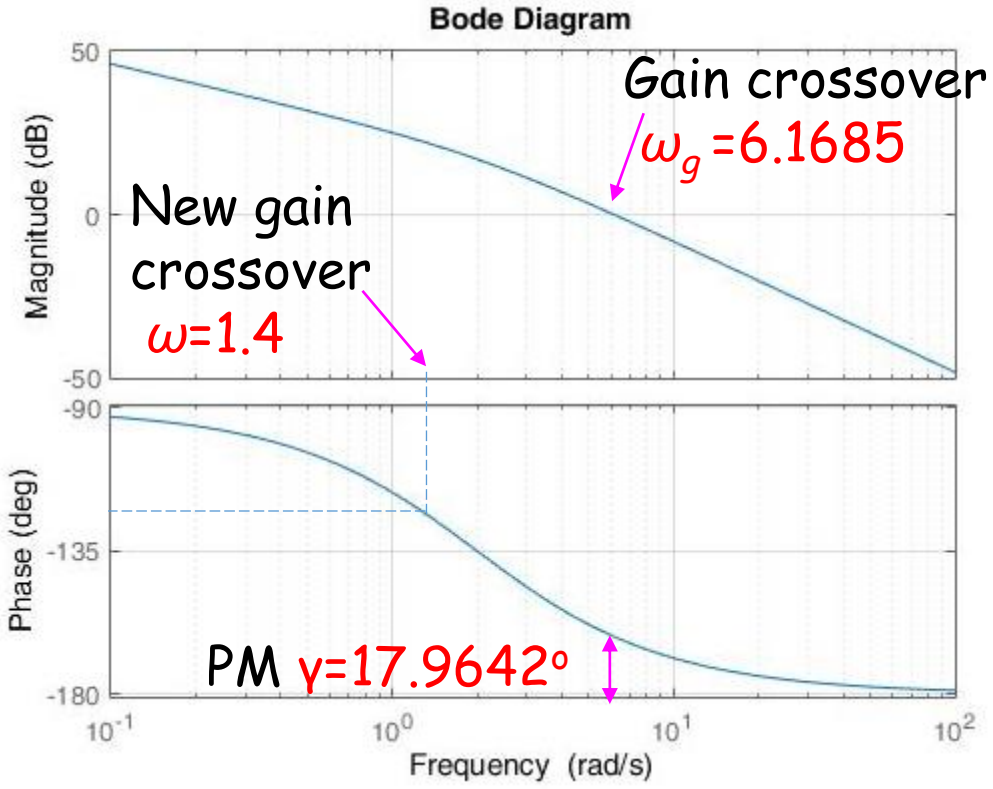
$$G_1(s) = KG(s) = \frac{40}{s(s+2)} \Rightarrow$$

$$G_1(j\omega) = \frac{40}{(j\omega)(j\omega+2)} = \frac{20}{(j\omega)(1+j\omega/2)}$$

Phase-Lag Compensator Design

Draw the Bode diagram of

$$G_1(j\omega) = \frac{20}{(j\omega)(1+j\omega/2)}$$



We need ϕ_d=50°. Choose ε=5°.

$$\phi = -180^o + 50^o + 5^o = -125^o$$

Find the frequency at which the phase angle of $G_1(j\omega)$ is -125°.

$$\angle G_1(j\omega) = -90^o - \tan^{-1}\left(\frac{\omega}{2}\right) = -125^o$$

$$\omega = 1.4 \text{ rad/sec}$$

This will be the new crossover frequency.

Phase-Lag Compensator Design

The magnitude at ω=1.4 is

$$20\log_{10}\left|\frac{20}{(j1.4)(1+j1.4/2)}\right| = 20\log_{10}\left(\frac{40}{1.4\times 2.44}\right) \Rightarrow \beta = \frac{40}{1.4\times 2.44} = 11.71$$

$$= 21.37 = -20\log_{10}\left(\frac{1}{\beta}\right)$$

Choose the zero of the compensator one decade below the new crossover frequency.

$$z = \frac{1.4}{10} = 0.14 = \frac{1}{\tau}$$

Choose the pole of the compensator

$$p = \frac{1}{\beta\tau} = \frac{0.14}{11.71} \cong 0.01$$

Determine the gain

$$K_c = \frac{K}{\beta} = \frac{10}{11.71} = 0.85$$

The compensator TF becomes

$$G_c(s) = 0.85\frac{s+0.14}{s+0.01}$$

Phase-Lag Compensator Design

The compensator TF is

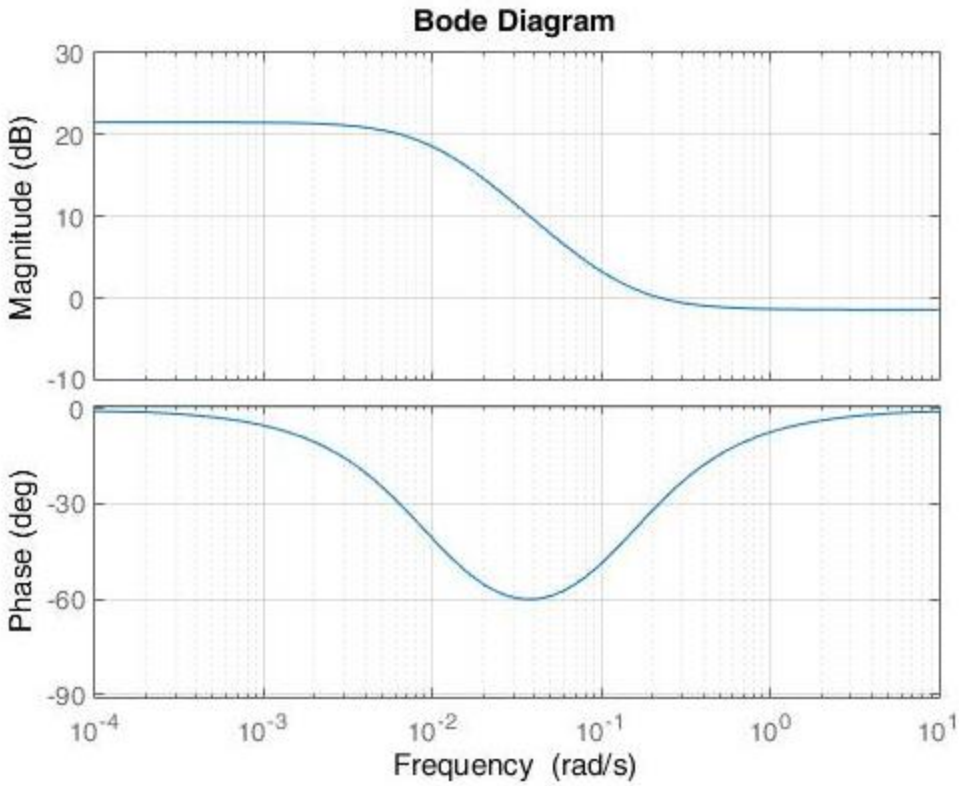
$$G_c(s) = 0.85\frac{s+0.14}{s+0.01}$$

At the new corner frequency the magnitude plot is negative in dB.

It shifts down the magnitude plot of the compensated system.

The added negative phase at low frequencies diminishes for high frequencies.

The Bode diagram of the compensator is



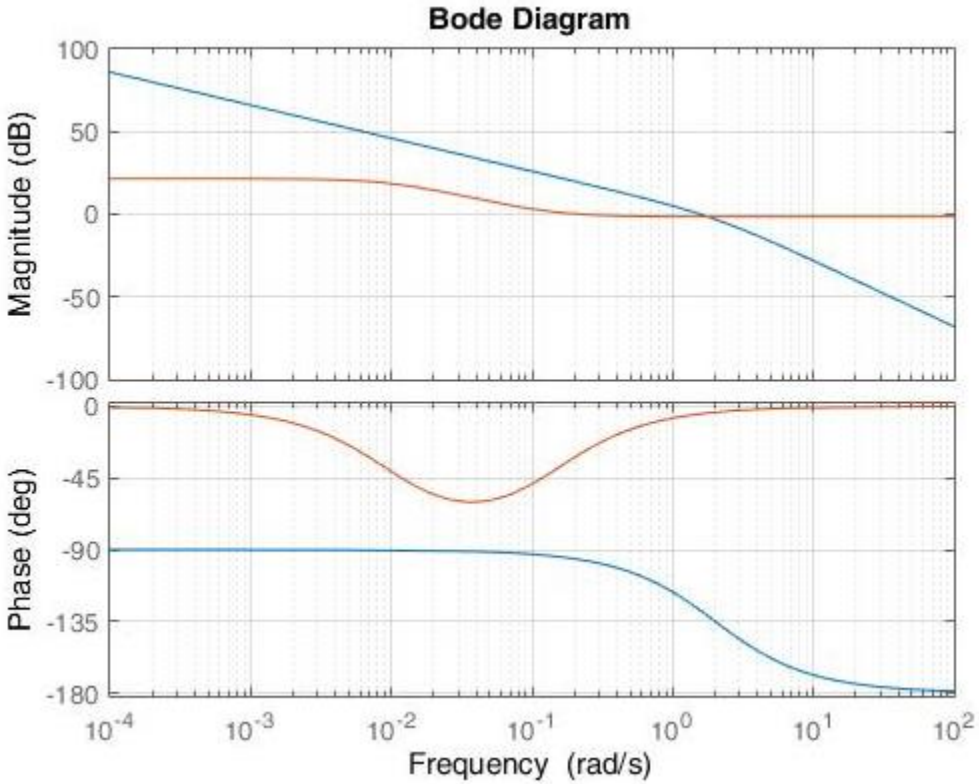
Phase-Lag Compensator Design

At high frequencies of interest the affect of the compensator is mostly in magnitude.

It shifts down the magnitude plot of the compensated system.

At very low frequencies the compensator contributes with positive magnitude in DB and negative phase.

The Bode diagrams of the plant and the compensator are



Phase-Lag Compensator Design

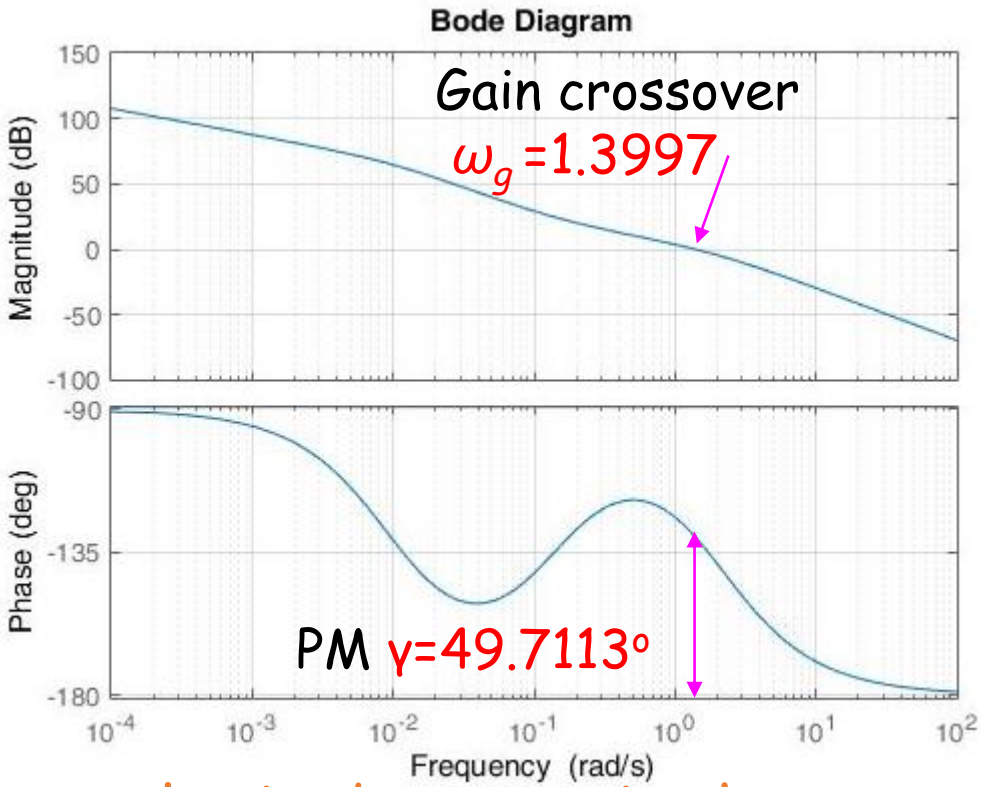
The OL TF of the compensated system is

$$G_c(s)G(s) = 0.85\frac{(s+0.14)}{(s+0.01)}\frac{4}{s(s+2)}$$

Note that for this system the gain margin is ∞.

Note that the bandwidth of the system decreased since we utilized a phase-lag compensator (the gain crossover frequency decreased from 6.1685 to 1.3997 and similarly the bandwidth).

The Bode diagram of the compensated system is



=> results in decrease in the speed of the response.

Since the specifications are satisfied, the design is completed.